

CampusEats

Verbs

- Browse Outlets
- View menu
- Add item to cart
- Place order
- Make payment
- Track order status
- Receive notification
- Collect/Pickup order
- Cancel order
- Rate/Review outlet or order
- Accept/Reject order
- Mark order as ready
- Manage outlets

Contract

1. Order Service

Operation	Input	Output	Errors
<code>addToCart</code>	studentId, itemId, quantity	cart with updated items	item unavailable, invalid quantity
<code>placeOrder</code>	studentId, items, paymentMethod	orderId, status, total, estimatedReadyMinutes	empty cart, item unavailable, payment declined
<code>getOrder</code>	orderId	order details, status	order not found
<code>cancelOrder</code>	orderId	cancellation status	order not found, too late to cancel
<code>rateOrder</code>	orderId, rating, review	review confirmation	order not found, already reviewed

2. Catalogue Service

Operation	Input	Output	Errors
<code>listOutlets</code>	filter	list of outlets	no outlets found
<code>getMenu</code>	outletId	menu items with prices	outlet not found
<code>checkItem</code>	outletId	availability, current price	item not found
<code>manageOutlet</code>	outlet information, action	updated outlet status	outlet not found, invalid data

3. User/Profile Service

Operation	Input	Output	Errors
<code>validateStudent</code>	studentId	valid/invalid, profile status	student not found, account inactive

4. Payment Service

Operation	Input	Output	Errors
<code>charge</code>	amount, paymentMethod	payment status, receipt	payment declined, invalid payment method
<code>refund</code>	payment reference, amount	refund status	payment not found, refund not allowed

5. Notification Service

Operation	Input	Output	Errors
<code>sendNotification</code>	studentId, event, message	notification status	invalid student, delivery failed

Central Operation: `placeOrder`

`placeOrder` is the central operation of the CampusEats system. It creates a food order after validating the student, checking the requested items, and completing payment.

Inputs

Input	Description
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<code>studentId</code>	Identifies the student placing the order
<code>items</code>	List of food items and requested quantities
<code>paymentMethod</code>	Payment method selected by the student

Outputs

Output	Description
<code>orderId</code>	Identifier of the newly created order
<code>status</code>	Current order status
<code>total</code>	Total amount charged
<code>estimatedReadyMinutes</code>	Estimated time until the order is ready for pickup

Error Cases

Error	Condition
<code>StudentNotFound</code>	Student account does not exist or is inactive
<code>ItemUnavailable</code>	One or more requested items are unavailable
<code>InvalidQuantity</code>	Requested quantity is invalid
<code>EmptyCart</code>	No items were provided for the order
<code>PaymentFailed</code>	Payment could not be completed
<code>OrderCreationFailed</code>	Order could not be created

Internal Details Hidden from Callers

The caller does **not** need to know how `placeOrder` works internally. The Order Service:

1. Validates the student through the **User/Profile Service**.
2. Checks each requested food item through the **Catalogue Service**.
3. Calculates the order total.
4. Requests payment through the **Payment Service**.
5. Creates the order and records its items and status.
6. Updates the order status to the appropriate initial state.
7. Sends an order confirmation through the **Notification Service**.
8. Returns the order result to the caller.

Validation Table

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled	Failing Property
User/Profile	✓	✓	✓	✓	✓	-
Catalogue	✓	✓	✓	✓	✓	-
Order	✓	✓	✓	✓	✓	-
Payment	✓	✓	✓	✓	✓	-
Notification	✓	✓	✓	✓	✓	-